Stockholm University

Department of Meteorology

Climate and the general circulation (MO7021)

The Arctic Climate

March 3, 2026

Author: Fredrik Bergelv (fredrik.bergelv@live.se)

Supervisor: Johan Nilsson

Contents

1 Introduction **2**

1.1 Background - The Arctic region 2

2 Data and Methodology **3**

2.1 Methodology 3

2.2 The data 4

3 A brief overview of the regional climate **8**

3.1 Things to write bout 8

4 Regional Climate and the General Circulation **9**

5 Climate Changes: Ongoing and Projected for the Future **9**

6 The Climate Impact on Societal Activities **9**

7 Conclusion **9**

1 Introduction

The Arctic and the Antarctic are two unique regions on Earth that are characterized by extreme cold temperatures, vast ice sheets, and unique ecosystems. The Arctic is located in the Northern Hemisphere and is centred around the North Pole. In this report, we will discuss the climate in the High Arctic together with the island of Greenland to develop a deeper understanding of how this region is affected by the general circulation, climate change, and social activities in the region. This will be done using articles and books on the topic together with reanalysed data from the region.

1.1 Background - The Arctic region

When dealing with the Arctic, it is important to define the region. The Arctic is commonly defined as the area north of the Arctic Circle, which is located at 66°33" N latitude [National Snow and Ice Data Center, 2026, Serreze and Barry, 2014]. However, others define the Arctic based on the extent of the Arctic ecosystem, which requires a July mean temperature under 10°[National Snow and Ice Data Center, 2026]. Using this definition, the Arctic can be divided into two main sub-regions, including the Arctic maritime region and the Arctic continental region according to National Snow and Ice Data Center [2026]. The important geographical locations mentioned in this report can be seen in Figure 1

The Arctic maritime region includes the Arctic Ocean and its surrounding seas such as the Bering and Greenland Seas, as well as the Labrador Sea and Baffin Bay. This region is sometimes described as a Mediterranean-type ocean due to its limited connection to the Atlantic Ocean and the Pacific Ocean [Danilov, 2000]. Looking at the bathymetry of the Arctic Ocean, one can observe that the Bering Strait, but also the Arctic Archipelago and the Barents Sea, are shallow with depths of hundreds of meters. Comparing this with the Fram Strait between Svalbard and Greenland, it is much deeper, allowing a bottleneck water flow between the Arctic and the Atlantic [Jakobsson et al., 2003]. Thus, it is very reasonable to view this ocean as a semi-isolated ocean, just like the Mediterranean.

The Arctic continental region consists of places like Svalbard, northern Canada, Russia, and the Nordic countries. However, the main landmass is Greenland, covering a large area of the Arctic. The Arctic is often split into the Low and High Arctic to distinguish between the more forested regions and the tundra up north [Bliss, 1997]. The landmass of Greenland is thus characterized by a tundra landscape, although along the northern coast it becomes more suitable to talk about a polar desert due to the lack of moisture [Charlier, 1969]. Furthermore, only 2–3% of the polar desert experiences some type of vegetation during the summer months [Bliss, 1997].

Figure 1: Arctic map showing the Arctic Circle (red dashed line) and some key locations mentioned in the text (marked as blue dots). Note that the Nordenskiöld Sea is also known as the Laptev Sea.

When describing the Greenlandic landmass, one can use the categorization developed by Wladimir Köppen [1884] to categorize the terrain. Köppen divided the world's landmasses into five main climate groups: tropical (A), arid (B), temperate (C), continental (D), and polar (E). Furthermore, each subgroup could be divided into more groups depending on precipitation and later on temperature. Due to the constantly cold nature of the Greenlandic landmass, the region was only divided into two main groups: the tundra (ET) and the ice cap (EF).

The Arctic ice cap is an important concept for understanding the Arctic region due to its high albedo and effect on the region. The ice cover in the Arctic is referred to as the cryosphere. On Greenland, the ice cap covers $1759\,\mathrm{km}^2$, which is about 81% of the total area [Serreze and Barry, 2014]. Arctic sea ice also plays a very important role in the region, as it turns the ocean into a land-like state. Furthermore, the melting and freezing of the ice are directly coupled with the atmospheric circulation and energy budget, as it ranges from $15 \times 10^6\,\mathrm{km}^2$ in March to $5 \times 10^6\,\mathrm{km}^2$ in September [Serreze and Barry, 2014]. The extent of the sea ice cover is determined by nearby landmasses, the inflow of warmer water through the Fram Strait, and drainage from large river systems such as the Mackenzie, Lena, Yenisey, and Ob [Serreze and Barry, 2014]. The rivers, together with the inflow of fresh water from the Pacific (freshness being a product of many river drainages in the region) and the region experiencing more precipitation than evaporation, lead to a low saline content, creating ideal conditions for ice formation. The sea ice is constantly in motion due to the Beaufort Gyre and Transpolar Drift, which transport the ice shelves, with some ice being transported through the Fram Strait. From there, the East Greenland Current carries cold Arctic water and sea ice into the North Atlantic, where it eventually melts.

2 Data and Methodology

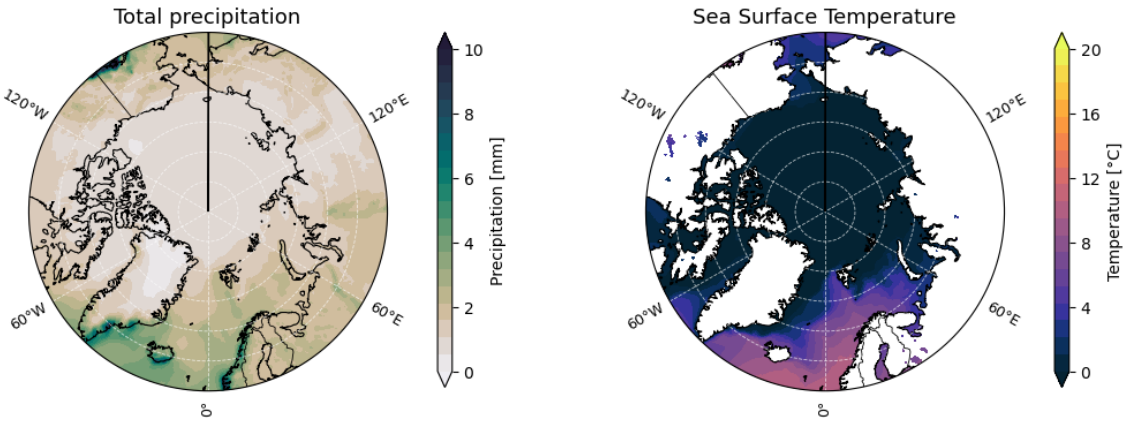
2.1 Methodology

The data used in this report is based on reanalysis data, which is a combination of model simulations and observational data. The data comes from the ERA5 reanalysis produced by the European Centre for Medium-Range Weather Forecasts (ECMWF) and obtained from the Copernicus Climate Data Website. ERA5 is a global atmospheric reanalysis that provides a physically consistent estimate of the state of the atmosphere by assimilating observational data into numerical weather prediction models [Hersbach et al., 2020]. This dataset has a horizontal resolution of $0.25^\circ \times 0.25^\circ$ and covers the period from 2012 to 2021. Monthly mean fields from 2012 until 2021 were analysed. The data were processed using Python (xarray), and graphical plots were produced using matplotlib and cartopy to visualize spatial patterns and seasonal variability in the Arctic region.

The reanalysis data has strengths in providing a comprehensive and continuous dataset that covers a long period of time and a wide range of variables. However, it also has limitations. The reanalysed data combine observations from various sources, including satellites, weather stations, aeroplanes, and buoys, however some regions, especially in the Arctic, have sparse observational coverage, which lead to a stronger dependence on the ones present such as satellites [Hersbach et al., 2020]. This can give rise to errors and biases such as measuring cloud cover, due to the white surface of the ice and snow leading to an overestimation of the cloud cover in the region [Hersbach et al., 2020]. Other more complex interactions, such the sea ice and atmospheric interaction, will then be heavily dependent on the model's ability to capture these interactions. Different thicknesses and coverage of sea ice will affect the atmospheric temperature, radiance, and albedo in that area and will thus directly affect the atmospheric circulation and energy budget. Due to the the complexity of measuring ice cover, and especially ice thickness, the data is mostly model based data. However the model only has a spatial resolution of $0.25^\circ \times 0.25^\circ$, which is not enough to capture small ice shelves in the ocean. For example, if a region has 50% sea ice cover, the model will only be able to capture this as one averaged value. Thus, the uncertainties from the model simulations can be higher due to lack of observations. Another important aspect to consider is temporal resolution. The data used in this report is based on monthly mean fields, which means that it may not capture short-term variability in the Arctic region. This can include phenomena such as storms or cyclones, which can have a significant impact on the atmospheric conditions in the region.

2.2 The data

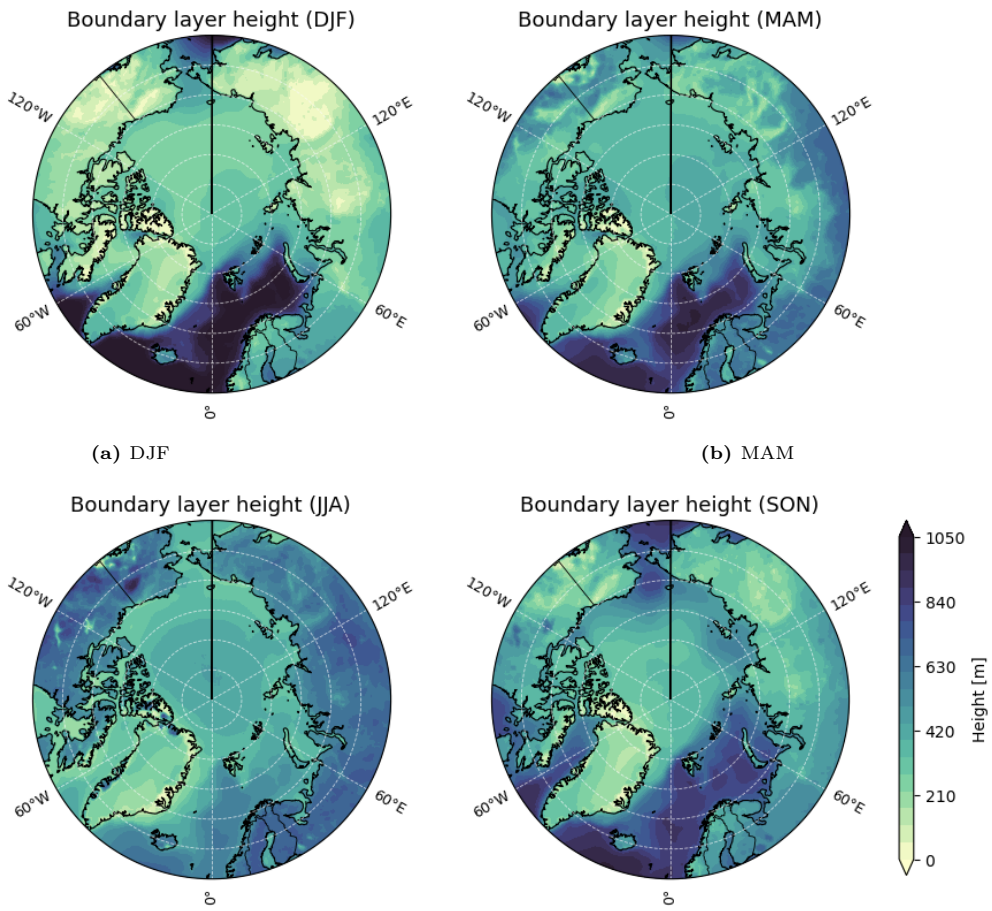
Above in the background section two main sub-regions of the Arctic were defined, the Arctic dessert and the Arctic tundra. It is therefore relevant to evaluate the precipitation in the region, as it is a key factor in determining the type of ecosystem present. The annual mean percipitation for the period 2012–2021 can be seen in Figure 2a. The figure shows that the Arctic region is generally dry, with most areas receiving less than 500 mm of precipitation per year. The highest precipitation levels are observed along the coastlines, particularly in the Barents Sea and the Bering Sea, where values can exceed 1000 mm per year. This is consistent with the fact that the Arctic region is characterized by low precipitation levels, which contribute to the formation of polar deserts and tundra ecosystems.



(a) Total Precipitation **(b)** Sea Surface Temperature

Figure 2: Annual mean total precipitation and sea surface temperature for 2012–2021.

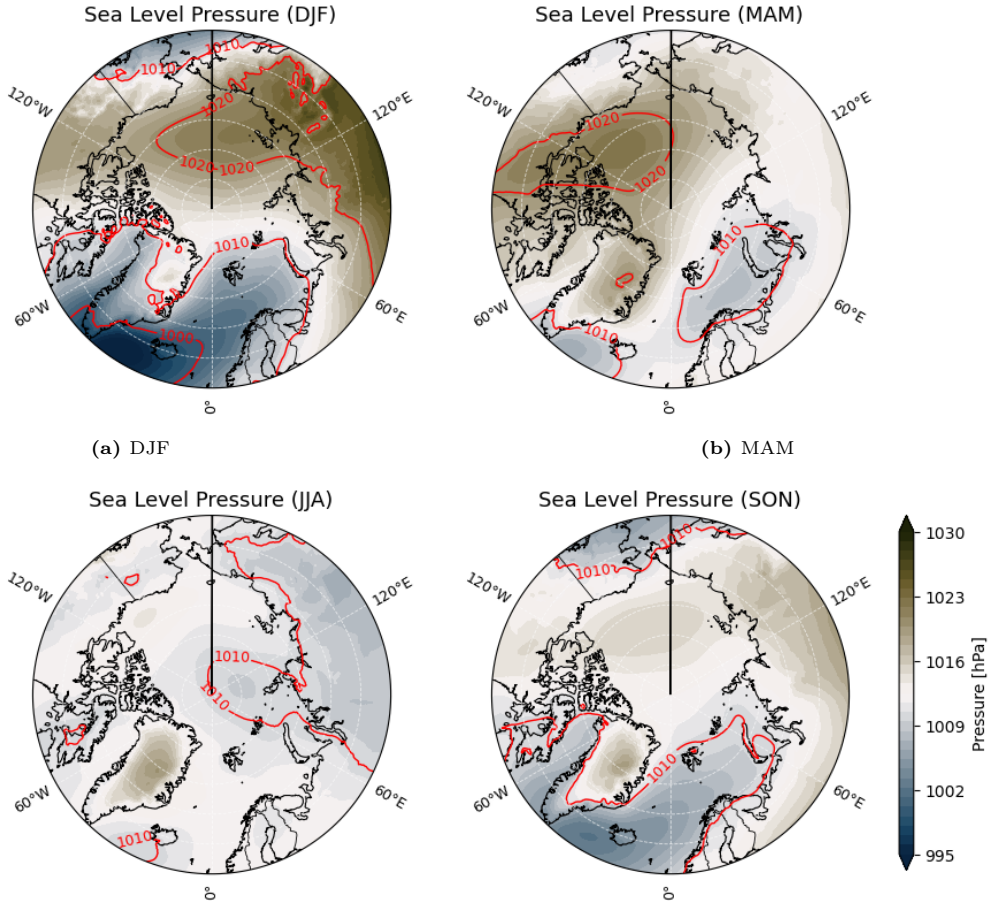
describe through which observations or techniques the data is constructed, as well as the methods you have used to analyse and present the data graphically. Here, it is important to discuss the reliability and limitations of the data as well as the analyses you have used.



(a) DJF **(b)** MAM

(c) JJA **(d)** SON

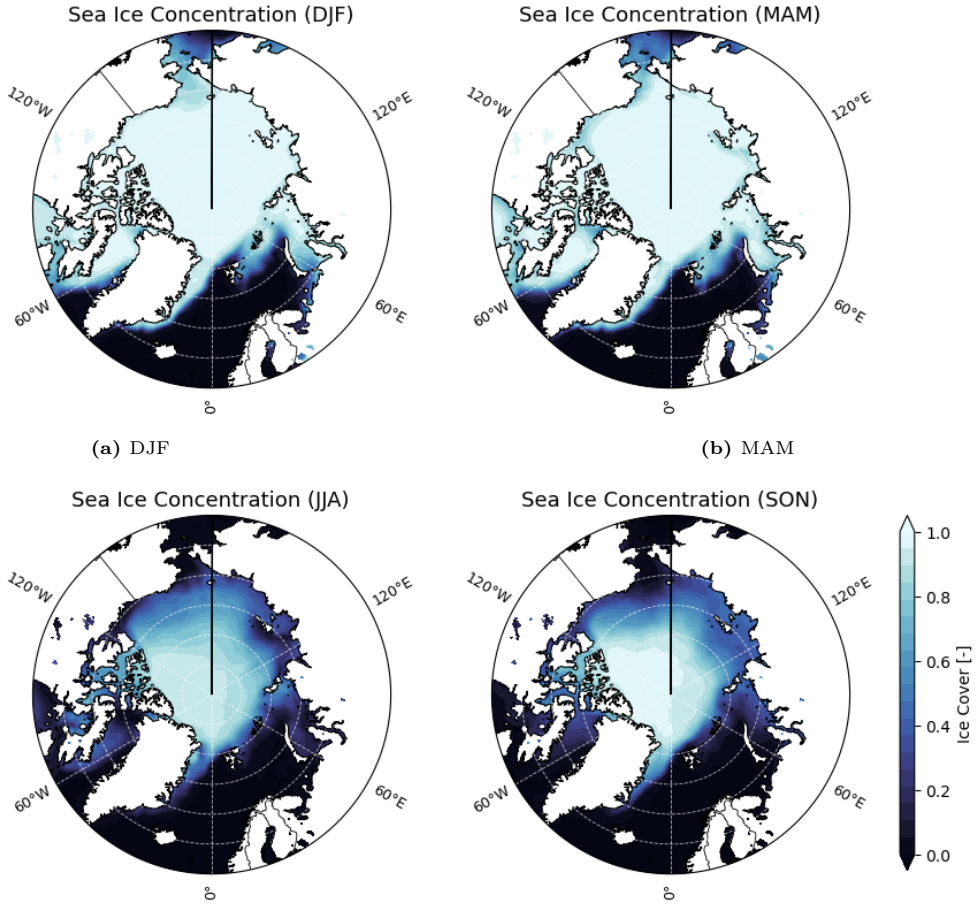
Figure 3: Seasonal mean boundary layer height (m) for 2012–2021 for DJF, MAM, JJA, and SON.



(a) DJF **(b)** MAM

(c) JJA **(d)** SON

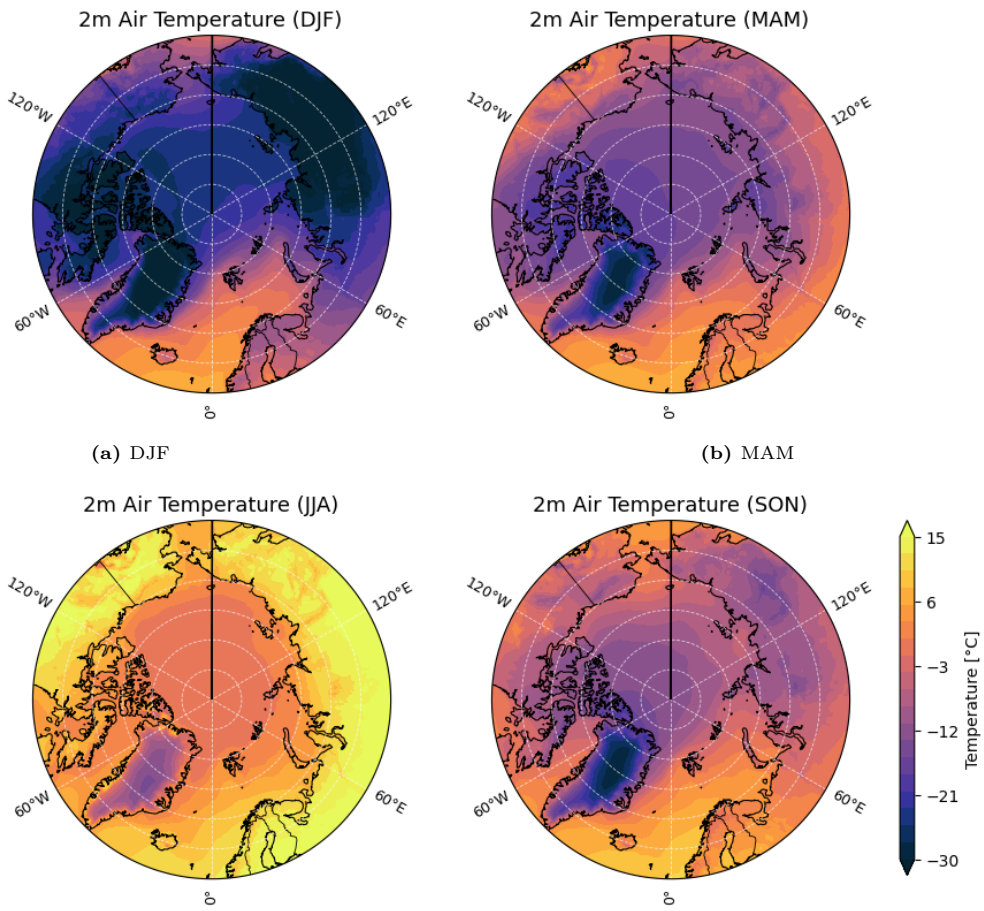
Figure 4: Seasonal mean sea level pressure (hPa) for 2012–2021 for DJF, MAM, JJA, and SON.



(a) DJF **(b)** MAM

(c) JJA **(d)** SON

Figure 5: Seasonal mean sea ice concentration (-) for 2012–2021 for DJF, MAM, JJA, and SON.



(a) DJF **(b)** MAM

(c) JJA **(d)** SON

Figure 6: Seasonal mean 2m air temperature (°C) for 2012–2021 for DJF, MAM, JJA, and SON.

3 A brief overview of the regional climate

Sea ice: part of the the hydrological cycle Maximum sea ice extent during the spring. Maximum during the spring equanox nad minimum autumn equanox.

3.1 Things to write bout

1. Köppen characteristics
2. Cryosphere
3. Geographical features, the atmospheric general circulation, regional weather patterns, and ocean currents
4. Based on the Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report (available online) account briefly for ongoing and future **climate change** in your study area
5. human society (ethical aspects)

4 Regional Climate and the General Circulation

Talk about the Hadley and Ferrel cell land how this creates a polar cell witch creates subsidence and and a polar dessert.

5 Climate Changes: Ongoing and Projected for the Future

6 The Climate Impact on Societal Activities

7 Conclusion

References

L.C. Bliss. *Polar and Alpine Tundra*, volume 3 of *Ecosystems of the World*. Elsevier, Amsterdam, 1997.

Charlier Charlier. The geographic distribution of polar desert soils in the northern hemisphere. *Geological Society of America Bulletin*, 80, 1969.

I.L. Danilov. *Arkticheskii okean kak faktor globalnykh klimaticheskikh izmenenii*. Nauchny Mir, 2000.

Hans Hersbach, Bill Bell, Paul Berrisford, Shoji Hirahara, András Horányi, Joaquín Muñoz-Sabater, Julien Nicolas, Carole Peubey, Raluca Radu, Dinand Schepers, Adrian Simmons, Cornel Soci, Saleh Abdalla, Xavier Abellan, Gianpaolo Balsamo, Peter Bechtold, Gionata Biavati, Jean Bidlot, Massimo Bonavita, Giovanna De Chiara, Per Dahlgren, Dick Dee, Michail Diamantakis, Rossana Dragani, Johannes Flemming, Richard Forbes, Manuel Fuentes, Alan Geer, Leo Haimberger, Sean Healy, Robin J. Hogan, Elías Hólm, Marta Janisková, Sarah Keeley, Patrick Laloyaux, Philippe Lopez, Cristina Lupu, Gabor Radnoti, Patricia de Rosnay, Iryna Rozum, Freja Vamborg, Sebastien Villaume, and Jean-Noël Thépaut. The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146 (730):1999–2049, 2020. ISSN 1477-870X. doi: 10.1002/qj.3803.

Martin Jakobsson, Arthur Grantz, Yngve Kristoffersen, and Ron Macnab. Physiographic provinces of the Arctic Ocean seafloor. *GSA Bulletin*, 115(12):1443–1455, December 2003. ISSN 0016-7606. doi: 10.1130/B25216.1.

National Snow and Ice Data Center. Arctic Weather and Climate. https://nsidc.org/learn/parts-cryosphere/arctic-weather-and-climate, 2026.

Mark C. Serreze and Roger G. Barry. *The Arctic Climate System*. Cambridge Atmospheric and Space Science Series. Cambridge University Press, New York, NY, USA, second edition edition, 2014. ISBN 978-1-107-03717-5.

Wladimir Köppen. Die Wärmezonen der Erde, nach der Dauer der heissen, gemässigten und kalten Zeit und nach der Wirkung der Wärme auf die organische Welt betrachtet. *Meteorologische Zeitschrift*, 20 (3), 1884.